

Evolution and management of the North American grassland herpetofauna

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Abstract.—The modern North American grassland herpetofauna has evolved in situ since the Miocene. Pleistocene glaciation had a minimal effect except in the far North, with only minor displacements of some species. South of the glaciers, winters were warmer and summers cooler than at present. Snake-like reptiles, leaping frogs, and turtle “tanks” are favored adaptive types in uniform dense grassland. A typical fauna consists of about 10-15 species, mostly snakes. Special habitat components, such as streams and ponds, bare ground, sand, trees, prairie dog towns, and rocky outcrops, add distinct suites of species. There is also an increase in species number from north to south and west to east. Grassland use and management, such as prairie dog control, off-road vehicle traffic, and brush removal, have demonstrable effects on the herpetofauna. However, the effects of three of the most widespread management procedures—water development, grazing, and fire—are largely unstudied. Although highly fragmented, the majority of species of grassland reptiles and amphibians are widespread and populations are resilient, but there are special conservation problems associated with Pleistocene relicts with limited distributions.

INTRODUCTION

At the time of the arrival of Europeans in North America, much of the interior of the continent was covered by grasslands. The heart of this great expanse was the tallgrass prairie of the midwestern United States. Where they were not bounded by mountain massifs, the lush grasslands feathered off to shortgrass prairies and desert grasslands in the west and south, shortgrass prairie and coniferous parklands and forests in the north, and deciduous forests in the east. The original grasslands and their accompanying faunas were continuously distributed, with few biogeographic barriers. With the westward expansion of European-style agriculture, most of the prairie was rapidly and completely converted to agriculture, producing a highly fragmented biota.

Parmenter et al. (1994) documented the exceptionally high vertebrate diversity in southwestern rangelands, and they emphasized that preservation of this biodiversity in the remaining habitat fragments will depend on skillful management of human activities in a fashion that integrates faunal biology, resource requirements, and historic movement patterns.

Nongame species, especially reptiles and amphibians, are not usually accorded the attention in land management decisions that their proportionate contribution to local biodiversity deserves. The situation has not changed much in the last 15 years (Bury et al. 1980); one of the few grassland management programs focused explicitly on reptiles or amphibians is the bullsnake (*Pituophis melanoleucus*) control trapping that has been carried out for decades at Valentine National Wildlife Refuge in Nebraska (Imler 1945). As a partial remedy, this review is intended to explore the characteristics of grassland herpetofaunas, to

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examine the issues involved in their management, and to make recommendations for managers concerned with enhancing the herpetological components of the remaining prairie fragments need to accomplish this important task. I will explore several areas:

- The history of the grassland herpetofauna
- The adaptations and habitats of grassland reptiles and amphibians
- The management of grassland reptiles and amphibians
- Conservation concerns
- Management recommendations

HISTORY OF THE GRASSLAND HERPETOFAUNA

Is there a coevolved North-American grassland herpetofauna?

Savage (1960) recognized a modern North American desert and plains herpetofauna whose boundaries are determined by zones of relatively rapid species transitions into different surrounding herpetofaunas (fig. 1). The present study deals with this herpetofauna, which inhabits the entire mid-continental grassland, including the Sonoran and Chihuahuan desert grasslands and the Great Basin. The grassland herpetofauna includes all of the reptiles and amphibians inhabiting this area, not just those that live in the grass. My conclusions are specific to the area; they often do not apply to grasslands outside of the area, such as the Central Valley of California, that have been long isolated from the grasslands in the central core of North America.

With the increasingly xeric climate of the late Miocene, the modern, unbroken grasslands began to form by coalescence of previously scattered and isolated fragments (Axelrod 1985). In marked contrast to mammalian faunas that experienced massive Pleistocene extinctions, North American herpetofaunas have changed remarkably little since the Pliocene. Pleistocene herpetofaunas from western North America were composed of most of the same species that are found there now (Holman 1995; Rogers 1982; Parmley 1990). The major difference stems from the loss of several

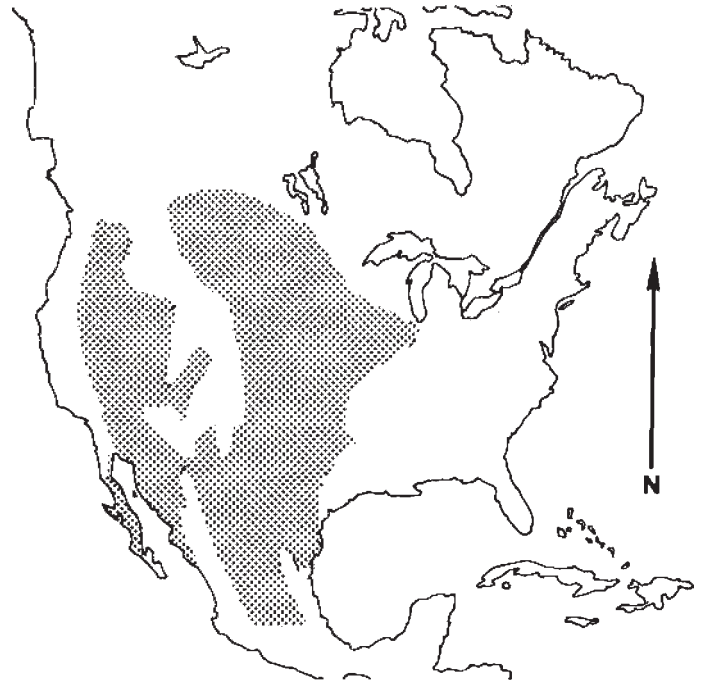


Figure 1. Distribution of the modern North American desert and plains herpetofauna. After Savage (1960).

tortoises of the genus *Geochelone* (Moodie and Van Devender 1979). Representatives of this ancient, worldwide radiation of medium to giant-sized tortoises were present in many fossil North American grassland faunas throughout the Tertiary. Members of the genus are still found in many parts of the tropics, but they are now extinct in North America. Based on well-documented fossil evidence, the Great Plains herpetofauna evolved in situ since at least the Miocene, with only minor east-west and north-south shifts that coincide with Pleistocene glaciations. Historic stability has produced a recognizable grassland herpetofauna that is relatively uniform across the North American plains.

The patterns of modern herpetofaunal diversity were accomplished mainly by the addition or deletion of species from a widespread suite of grassland forms. Most species are wide-ranging, supporting the notion that the grassland fauna is fairly homogeneous. For example, half (6 of 12) of the reptile and amphibian species found in the grasslands of Alberta and almost three-fourths (32 of 43) of the Kansas tallgrass species are also found in the grasslands of Chihuahua, several hundred kilometers to the south (Morafka 1977; Collins 1993; Russell and Bauer 1993).

What were the ecological conditions under which the herpetofauna evolved?

The presence of *Geochelone* in the fossil record of the midwestern United States from the Oligocene through the last glaciation (about 10,000 years ago) is especially enlightening (Moodie and Van Devender 1979). At the same time that *Geochelone* became extinct in North America, the ranges of four species of grassland tortoises of the genus *Gopherus* contracted into western, eastern and southern enclaves, where they survive to the present day (fig. 2). Moodie and Van Devender (1979) attributed these extinctions and range reductions to post-Pleistocene climatic deterioration, perhaps assisted by human predation. The fossil presence in Kansas of *Geochelone*, which is not known to dig burrows, was taken to indicate a climate that was essentially frost-free. Later, *Geochelone* died out, whereas *Gopherus* has survived at the southern and western edges of its Pliocene distribution probably because of its ability to escape predators and freezing temperatures by burrowing (Holman 1971, 1980; Moodie and Van Devender 1979; Rogers 1982).

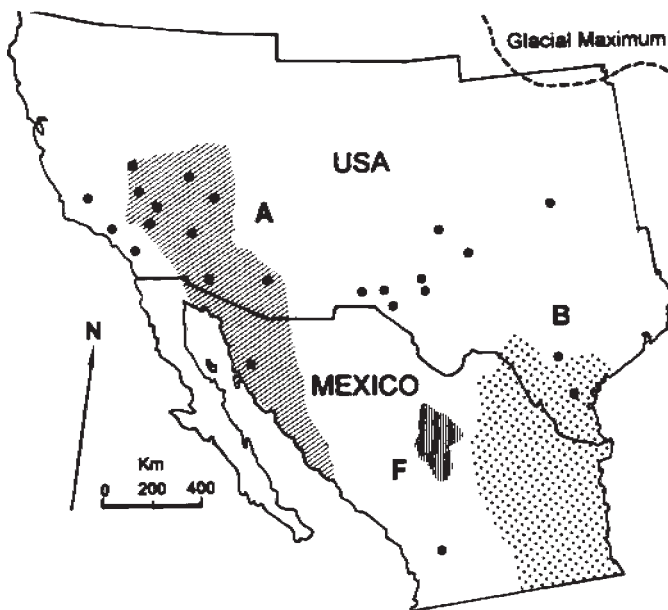


Figure 2. Modern distributions of three gopher tortoises (*Gopherus*; hatching and stipple) and late Pleistocene fossil localities for *Gopherus* and *Geochelone* (dots). After Moodie and Van Devender (1979).

The combined paleontological evidence from vertebrates, plants, and mollusks indicates that climates with cooler summers, milder winters, and possibly more precipitation prevailed in unglaciated North America until modern climatic patterns were established about 10,000 years ago (Holman 1995; Wright 1987). Climates south of the glaciers were relatively immune to the alternating glacial and interglacial cycles (Holman 1995). Bryson and Wendland (1967) hypothesized that the immense Pleistocene glaciers prevented the southward incursion of Arctic air masses that now bring sub-zero temperatures to most of the continent, and Rogers (1982) attributed the cooler summers and warmer winters to a greenhouse effect with heavy cloud cover and increased moisture. Under these conditions, the grassland herpetofauna has persisted largely intact to the present day.

ADAPTATIONS AND HABITATS OF GRASSLAND REPTILES AND AMPHIBIANS

What are the adaptive characteristics that enable reptiles and amphibians to live in grasslands?

We can gain some insight into the adaptations that allow reptiles and amphibians to inhabit grasslands by looking at their common characteristics. This is admittedly a speculative, circular process (if a grassland species has a trait, then that trait is an adaptation to grasslands), but some of the speculation will be useful, especially in those cases where we can identify a context in which the "adaptation" could operate.

The most common adaptive traits seen in grassland reptiles are leglessness and serpentine locomotion as seen in snakes and smooth, legless (or small-limbed) lizards (table 1). Legs are a liability when a serpentine reptile is threading its way between dense grass stems, and lizards that depend on their legs for locomotion are most common in grasslands where there are openings, such as rocky outcrops, tree trunks, or bare ground. As we shall see in the next section, the adaptive type exemplified by snakes appears to be very successful at exploiting uniform grasslands.

Frog locomotion seems to be another good way to negotiate grasslands. Grassland frogs have

Table 1. Adaptive types of reptiles and amphibians present in various North American grasslands. Figures in the table are the number of species and (percentages) of each adaptive type in the total terrestrial grassland herpetofauna in that area. "Toads" includes all dry, short-legged, hopping anurans that are resistant to desiccation (Bufonidae, Leptodactylidae, Microhylidae, Pelobatidae); "Frogs" includes moist, long-legged, leaping anurans that dry up easily (Ranidae, Hylidae); and "Snakes" includes snakes and legless lizards (Anguidae) that move by serpentine locomotion.

Adaptive type	Tallgrass Kansas ¹	Great Basin Western Utah ²	Southern Desert Chihuahua, Coahuila ³	Northern Southern Alberta ⁴
Salamanders	1 (2.3)	0 (0.0)	1 (1.8)	1 (8.3)
"Toads"	7 (16.3)	2 (9.5)	9 (16.4)	3 (25.0)
"Frogs"	4 (9.3)	1 (4.8)	1 (1.8)	2 (16.7)
Turtles	2 (4.7)	0 (0.0)	4 (7.3)	0 (0.0)
Lizards	7 (16.3)	9 (42.9)	18 (32.7)	1 (8.3)
"Snakes"	22 (51.2)	9 (42.9)	22 (40.0)	5 (41.7)
# of Species	43	21	55	12

¹ Collins 1993

² Stebbins 1985

³ Morafka 1977

⁴ Russell and Bauer 1993

exceptionally powerful, high-arched leaps that enable them to clear the top of the grassy barrier. A third successful type of locomotion in dense grasslands is "tank" locomotion, by which turtles are able to force their way through small gaps between grass clumps.

Many bodies of water in grasslands are ephemeral, generally filling in the winter and spring, and drying during the summer and fall. Therefore, many of the most successful amphibians are species that have relatively short aquatic larval stages.

What characteristics of grassland habitats are correlated with reptile and amphibian species diversity?

The tallgrass prairie of the eastern Great Plains is the simplest and most uniform of the grassland habitats. About 15 reptile species (mostly snakes) can be present in an undifferentiated tallgrass prairie in Kansas (table 2). Long-limbed lizards, several toad species, and a snake are able to use the grasslands in places where bare ground gives them the freedom to move about rapidly, allowing them to forage and escape effectively. The simple addition of temporary water sources can add up to seven species of amphibians, a turtle and three

snakes. Sandy or other loose, friable soil is necessary for species that dig. Rocky soils provide hiding places for small snakes, and rocky outcrops and trees provide elevated perches where collared (Crotaphytus) and prairie lizards (Sceloporus) can display. Rivers and large streams and their associated riparian zones finger out into the Great Plains from the east and south, bringing to the grasslands a series of frogs, turtles, and snakes that have their centers of distribution in the eastern United States.

As one moves west from Kansas tallgrass prairie toward the Great Basin, the grasslands become more open, tending to favor lizard species. On the other hand, they become drier, which diminishes the number of snakes, turtles, and amphibians. Total herpetofaunal species richness is halved, but the number of lizard species actually increases (table 1). Southward into the Chihuahuan Desert grasslands, the percentage of lizard species doubles and frogs decline, but overall species richness increases by about 25% with the addition of tropical "toads," turtles, and lizards (Morafka 1977; Van Devender and Bradley 1994). Northward into Alberta, species richness decreases dramatically, but the percent composition of different adaptive types remains roughly similar to that in Kansas.

Table 2. List of reptiles and amphibians in Kansas grasslands and their habitat correlates. Data from Collins (1993).

Common name	Scientific name	Common name	Scientific name
Inhabitants of undifferentiated grasslands		Inhabitants of sandy soils	
Ornate box turtle	<i>Terrapene ornata</i>	Glossy snake	<i>Arizona elegans</i>
Great Plains skink	<i>Eumeces obsoletus</i>	Western hognose snake	<i>Heterodon nasicus</i>
Prairie skink	<i>Eumeces septentrionalis</i>	Inhabitants of sandy soils and water	
Western slender glass lizard	<i>Ophisaurus attenuatus</i>	Woodhouse's toad	<i>Bufo woodhousei</i>
Racer	<i>Coluber constrictor</i>	Inhabitants of trees or rocks	
Western rattlesnake	<i>Crotalus viridis</i>	Eastern collared lizard	<i>Crotaphytus collaris</i>
Ringneck snake	<i>Diadophis punctatus</i>	Prairie lizard	<i>Sceloporus undulatus</i>
Night snake	<i>Hypsiglena torquata</i>	Great Plains rat snake	<i>Elaphe guttata</i>
Prairie kingsnake	<i>Lampropeltis calligaster</i>	Milk snake	<i>Lampropeltis triangulum</i>
Common kingsnake	<i>Lampropeltis getula</i>	Ground snake	<i>Sonora semiannulata</i>
Texas blind snake	<i>Leptotyphlops dulcis</i>	Plains black-headed snake	<i>Tantilla nigriceps</i>
Gopher snake	<i>Pituophis melanoleucus</i>	Inhabitants of rocky canyons and water	
Long-nosed snake	<i>Rhinocheilus lecontei</i>	Red-spotted toad	<i>Bufo punctatus</i>
Massasauga	<i>Sistrurus catenatus</i>	Inhabitants of rivers, riparian zones, or permanent ponds	
Lined snake	<i>Tropidoclonion lineatum</i>	Cricket frog	<i>Acris crepitans</i>
Inhabitants of temporary water		Bullfrog	<i>Rana catesbeiana</i>
Tiger salamander	<i>Ambystoma tigrinum</i>	Snapping turtle	<i>Chelydra serpentina</i>
Red-spotted toad	<i>Bufo punctatus</i>	Painted turtle	<i>Chrysemys picta</i>
Great Plains narrowmouth toad	<i>Gastrophryne olivacea</i>	Red-eared slider	<i>Trachemys scripta</i>
Spotted chorus frog	<i>Pseudacris clarki</i>	Smooth softshell turtle	<i>Trionyx muticus</i>
Boreal chorus frog	<i>Pseudacris maculata</i>	Spiny softshell turtle	<i>Trionyx spiniferus</i>
Western chorus frog	<i>Pseudacris triseriata</i>	Eastern hognose snake	<i>Heterodon platirhinos</i>
Plains leopard frog	<i>Rana blairi</i>	Plain belly water snake	<i>Nerodia erythrogaster</i>
Yellow mud turtle	<i>Kinosternon flavescens</i>	Diamondback water snake	<i>Nerodia rhombifer</i>
Checkered garter snake	<i>Thamnophis marcianus</i>	Northern water snake	<i>Nerodia sipedon</i>
Plains garter snake	<i>Thamnophis radix</i>	Graham's crayfish snake	<i>Regina grahamii</i>
Common garter snake	<i>Thamnophis sirtalis</i>	Brown snake	<i>Storeria dekayi</i>
Inhabitants of bare ground		Western ribbon snake	<i>Thamnophis proximus</i>
Six-lined racerunner	<i>Cnemidophorus sexlineatus</i>	Inhabitants of bare ground and water	
Lesser earless lizard	<i>Holbrookia maculata</i>	Great Plains toad	<i>Bufo cognatus</i>
Texas horned lizard	<i>Phrynosoma cornutum</i>	Green toad	<i>Bufo debilis</i>
Coachwhip	<i>Masticophis flagellum</i>	Plains spadefoot toad	<i>Spea bombifrons</i>

THE MANAGEMENT OF GRASSLAND REPTILES AND AMPHIBIANS

What management tools are available and what are their effects on grassland reptiles and amphibians?

Given the common components of all of the grassland herpetofaunas, general management principles often can be applied over wide areas (Table 3).

Water management

Water manipulation for the enhancement of livestock grazing and agriculture is ubiquitous in grasslands. Devices used range from bulldozer scrapes and check dams that temporarily store runoff and extend the grazing season, to complex

systems that include permanent reservoirs behind high dams, ditches, channeled streams, and eradication of phreatophytes (Elmore and Kauffman 1994).

All grassland amphibians require water for the development of their larvae. The great majority of species use temporary ponds that hold water for at least several months in spring and summer and not streams or rivers. At one time, buffalo wallows probably provided temporary pond habitats in many grasslands, but these have largely disappeared. The suitability of a pond for amphibians is tightly tied to the duration of its water cycle: If ponds dry up too soon, larvae will die; if they last too long, predatory mammals, aquatic insects, and even fish will become abundant.

Ponds that contain water much of the year attract mammalian predators such as opossums (*Didelphis*), coyotes (*Canis*), foxes (*Vulpes*), badgers (*Taxidea*), skunks (*Mephitis*), and raccoons (*Pro-*

Table 3. Effects of management techniques on grassland herpetofaunas.

Management practice	Positive effects	Negative effects
Creation of temporary ponds ¹	Habitat for native amphibians and mud turtle	Increase in mammalian predators
Provision of permanent water ²	Habitat for aquatic turtles, snakes, bullfrogs	Addition of many terrestrial and aquatic vertebrate predators
Moderate grazing ³	Increases lizard species diversity?	Unknown
Heavy grazing ³	Some lizards densities increase	Some lizards decrease; some lizard species lost
Brush removal ⁴	Some lizard populations increase	Most lizards decline, some species lost, tortoises decline
Grass planting ⁵		Most lizards decline, some species lost?
Burning ⁶		Box turtles burned
Prairie dog control ⁷		Rattlesnakes, earless lizards decline
Off-road vehicles ⁸		Fewer lizards

¹ Personal observation

² Moyle 1973, Hayes & Jennings 1986, Schwalbe & Rosen 1989

³ Jones 1981

⁴ Busack and Bury 1974, Berry 1978, Germano and Hungerford 1981, Rose and Judd 1982, H.L. Snell (personal communication)

⁵ The effects of exotic grass introductions on herpetofaunas have not been studied, but Bock et al. (1983) and D'Antonio and Vitousek (1992) discuss impacts on other elements of the grassland biota

⁶ Bigham et al. 1965, Erwin and Stasiak 1979

⁷ Klauber 1972, Miller et al. 1990

⁸ Busack and Bury 1974, Vollmer et al. 1976, Bury et al. 1977

cyon). Christiansen and Gallaway (1984) increased the hatchling recruitment of seven species of turtles three to four fold by removing raccoons from the vicinity of an Iowa pond. Recruitment declined again after raccoons became reestablished.

Ponds that hold water the entire year, year after year, have ecological consequences distinct from those of temporary ponds. Permanent water provides habitat for the bullfrog (*Rana catesbeiana*) plus a number of fishes, snakes, and turtles that usually only penetrate the grasslands along permanent streams and rivers. The introduction of fish, especially predatory centrarchids and catfish, and bullfrogs has been implicated in the disappearance of native species in several places (Moyle 1973; Hayes and Jennings 1986; Schwalbe and Rosen 1989; Collins et al. 1989).

Ponds that enhance grasslands for amphibians and reptiles need not be sophisticated. A simple scrape that collects and holds spring rain water and snow melt through July is sufficient for many species. Ponds that are fenced from livestock are often more attractive to reptiles and amphibians because they usually have a better cover of emergent aquatic and riparian plants. Cows also degrade the banks and eventually churn a shallow pond into a muddy morass (personal observation).

Livestock grazing

Livestock grazing is the most widespread human-mediated influence on grassland vegetation in the American West (Fleischner 1994 and references therein). In addition to the direct effects of grazing, management of grasslands for cattle often includes brush and tree clearing and planting exotic grasses. Vegetative structure has been shown to affect North American lizard communities in predictable ways (Pianka 1966; Germano and Lawhead 1986), but nothing is known about the responses of other reptiles and amphibians. In many North American grasslands and savannas, grazing has been implicated in the invasion of woody vegetation and non-native grasses; in others, especially riparian zones, shrubs and trees are suppressed or eliminated by grazing livestock (D'Antonio and Vitousek 1992; Vavra et al. 1994 and references therein). The removal of riparian

vegetation and the mechanical destruction of stream and pond banks by cows and sheep can have a significant detrimental effect on the herpetofauna (Szaro et al. 1985; Jones 1989).

In dense grasslands, moderate grazing can increase vegetative patchiness and structure, opening up areas for use by widely-foraging lizards, snakes, and even toads. In drier, more open grassland formations, grazing effects are less predictable. Heavy grazing can reduce patchiness and promote homogeneity, which may benefit a few species of essentially desert lizards (Busack and Bury 1974), but it can also facilitate the invasion of brush and trees, which provide habitat for arboreal species at the expense of widely-foraging terrestrial lizards (Jones 1981). In southern Texas, heavy grazing favors the growth of prickly pear (*Opuntia*) which is a staple in the diet of the Texas tortoise (*Gopherus berlandieri*, Auffenberg and Weaver 1969). Claims that competition for food and mechanical injury by livestock have had a detrimental effect on Mojave Desert tortoise (*Gopherus agassizii*) populations may be true (Woodbury and Hardy 1948; Berry 1978; Coombs 1979; Nicholson and Humphreys 1981; Campbell 1988), but they are disputed (Bostick 1990) and have not been tested (Oldemeyer 1994).

Brush removal

Brush removal simplifies habitat structure, benefiting some reptile and amphibian species and eliminating others. The removal of competing shrubs allows grass to proliferate. Rose and Judd (1982) believe that brush clearing in southern Texas is detrimental to tortoise populations, but Auffenberg and Weaver (1969) think that the damage is short-term, and that tortoise habitat is ultimately enhanced by the practice because grass is encouraged. Partial removal of mesquite (*Prosopis juliflora*) in an Arizona grassland changed the proportions of the lizard species, but the species present and total numbers remained about the same; complete mesquite removal reduced the number of lizards by more than 50% and three common species virtually disappeared (Germano and Hungerford 1981). Increased grass density on the mesquite-free plot was suggested as the reason for the disappearance of some of the widely-

foraging lizard species. In a depauperate reptile fauna in Great Basin grasslands of eastern Oregon, Werschkul (1982) found lizards in all habitats with shrubs or rocky outcrops; he saw no lizards in grasslands that had been "restored" and lacked woody vegetation. In New Mexico, populations of the dune lizard (*Sceloporus arenicolus*) were reduced by 70-94% in areas where shinnery oak (*Quercus harvardii*) was removed (H.L. Snell personal communication). Invasion of the treated areas by dense stands of grasses and forbs was thought to be the main reason for the disappearance of the dune lizards.

Grass plantings

Aside from the accidental introduction of exotic grasses into sensitive native grasslands, at least three invasive, non-native perennial grasses are being planted as livestock forage in different parts of the North American grasslands (D'Antonio and Vitousek 1992). The effects of these plantings on the herpetofauna have not been studied in any detail, but many lizards, especially *Uta*, *Holbrookia* and *Cnemidophorus*, seem to be sensitive to grass clogging their foraging, display, and escape pathways (Germano and Hungerford 1981 personal observation).

Eurasian crested wheatgrass (*Agropyron desertorum*) has been widely planted throughout the Great Basin as a drought-tolerant replacement for the native *A. spicatum* (D'Antonio and Vitousek 1992, Rogler and Lorenz 1983).

Buffelgrass (*Cenchrus ciliaris*) is planted for cattle forage in the lower Rio Grande Valley of southern Texas and northeastern México, and in the Sonoran Desert (Nabham 1994; Warren 1994). The grass is very successful at colonizing open ground between mesquite and ironwood (*Olneya tesota*) trees. Two species of lizards that depend on open spaces and running for foraging and escape, the reticulate collared lizard (*Crotaphytus reticulatus*) and the spot-tailed earless lizard (*Holbrookia lacerata*), are restricted to the lower Rio Grande Valley. Buffelgrass clogs their habitat, rendering it unsuitable (A.H. Price personal communication).

A similar, and perhaps more widespread, problem is occurring with the widespread use of African Lehmann's lovegrass (*Eragrostis*

lehmanniana) in many parts of southeastern Arizona and southern New Mexico. Lovegrasses have been shown to diminish or eliminate many native populations of plants, birds, rodents, and grasshoppers (Bock et al. 1986).

Burning

Herpetofaunal responses to burning of North American grasslands have scarcely been studied. Box turtles (*Terrapene ornata*, *T. carolina*) seem to suffer fairly heavy losses in some situations, but other reptile populations are hardly affected. After an August fire on a 160 ha pasture in eastern Oklahoma, Bigham et al. (1965) found 25 dead and three live box turtles, a dead rattlesnake (*Crotalus* sp.), a dead copperhead (*Agkistrodon contortrix*), a live black rat snake (*Elaphe obsoleta*), and a live coachwhip snake (*Masticophis flagellum*). Erwin and Stasiak (1979) looked for dead animals in a 65 ha restored prairie in three successive springs after controlled burns. They found dead snakes after one of the fires: a bullsnake and two garter snakes (*Thamnophis radix*, *T. sirtalis*). Four live snakes were also seen. Legler (1960) assumed that ornate box turtles (*Terrapene ornata*) were occasionally killed by prairie fires, but he found no carcasses after an early April fire on his study plot. Two years later, he did find a large female box turtle that had been badly scarred by fire.

Mushinsky (1985) found that regular fires in Florida sandhills increased herpetological diversity and abundance, especially populations of the six-lined racerunner (*Cnemidophorus sexlineatus*). The intensity, timing, and frequency of burning help determine the structure of the vegetation, and lizard communities, especially, should be affected. Bury and Smith (1986) recommended small control burns to open up shrub grasslands for Texas tortoises in south Texas.

Prairie dog control

Because prairie dogs (*Cynomys* spp.) theoretically compete with livestock for forage, their eradication is a time-honored management practice among western ranchers (Miller et al. 1990). Eradication has been very effective; it is estimated

that prairie dogs persist on only about 2% of their former range (Ceballos et al. 1993; Miller et al. 1994). Nonetheless, a recent cost/benefit analysis concluded that prairie dog control programs operate at a net economic loss, even if the effects on non-target species, such as the black-footed ferret (*Mustela nigripes*), or the prairie dog's role in soil formation and nutrient cycling are not considered (Collins et al. 1984; Miller et al. 1990; Sharps and Uresk 1990; Samson and Knopf 1994).

Prairie dogs are "keystone" species in their ecosystem, and their towns have been called species diversity oases on the arid plains (Miller et al. 1990, 1994). Some species, such as the ferret, absolutely depend on prairie dogs, and others, including many reptiles and amphibians, are strongly associated with prairie dog towns. Twelve amphibian and 17 reptile species have been recorded from black-tailed prairie dog (*C. ludovicianus*) towns, and the lists do not appear to be exhaustive (Reading et al. 1989, Sharps and Uresk 1990). Where rocky outcrops are not available, prairie dog burrows may provide the only winter retreats for western rattlesnakes (*Crotalus viridis*) in northern sections of the Great Plains (Klauber 1972). Other reptiles and amphibians probably use prairie dog burrows as both summer and winter refuges and for nesting. Although no studies have been done, the elimination of prairie dogs may cause the local extinction of dependent segments of the herpetofauna.

Off-road vehicles

The effects of off-road vehicles are becoming an increasing concern in many grassland habitats, especially in drier areas that are near large population centers (Bury et al. 1977, Webb and Wilshire 1983). Busack and Bury (1974), Vollmer et al. (1976), and Bury et al. (1977) documented mild to drastic reductions in the species numbers and abundance of reptiles on Mojave Desert plots subjected to various degrees of off-road vehicle use. Severe off-road vehicle damage to arid grasslands is largely irreversible in any reasonable time frame, and unless managers can devise more effective ways of dealing with the problem, the rate of destruction will accelerate.

CONSERVATION CONCERNS

With few exceptions, the reptile and amphibian species associated with North American grasslands are widespread, common, and resilient, and they adapt to most perturbations, short of urbanization or crop farming. In the past, the large-scale conversion of grasslands to cropland has been the most damaging force, but no species have been lost on a global scale. Now that the grassland conversion process is mostly complete, the greatest danger to the herpetofauna is the cumulative effect of many small perturbations. Some of these (e.g., highway construction, urbanization, off-road vehicles, campground development) are almost always destructive, but others (e.g., check dams on small streams, irrigation schemes, land drainage, brush and tree invasion) may favor some elements of the herpetofauna, usually at the expense of other segments. These numerous, small-scale disturbances are difficult to monitor. An even more difficult challenge will be convincing management agencies that these perturbations need to be evaluated to see if they should be minimized or eliminated (Parmenter et al. 1994). In the aggregate, however, these habitat losses can spell the doom of a local fragment of a once cohesive herpetofauna.

The widespread continuing degradation of grasslands used as rangelands is a seemingly intractable problem. Many native grassland types are in danger of extinction by replacement with exotic stands. Management should focus on maintaining patches of sufficient size to provide refugia for lizards, such as *Cnemidophorus inornatus*, *C. uniparens*, and *C. neomexicanus*, that are thought to be dependent on these native plant formations. We need detailed information on the implications to the herpetofauna of changing the structure and species composition of grasslands by exotics like buffleggrass and Lehmann's lovegrass (W.G. Whitford personal communication).

Most of the species in the grassland herpetofauna are still widespread and secure; however, there are several areas that harbor relict species with very restricted ranges such as the dune lizard, the reticulate collared lizard, and the spot-tailed earless lizard described above (table 4). Many states protect species that barely enter their borders (e.g., species of the "Prairie Peninsula" of

Table 4. Areas of North American grasslands that support threatened species, subspecies, and populations of reptiles and amphibians.

Area	Species	Current threats
Mexican plateau	Bolson tortoise <i>Gopherus flavomarginatus</i>	Human predation ¹
South Texas, northeastern México	Reticulate collared lizard <i>Crotaphytus reticulatus</i> Spot-tailed earless lizard <i>Holbrookia lacerata</i> Black-spotted newt <i>Notophthalmus meridionalis</i>	Invasion of exotic grass, agriculture ²
Laramie Basin, Wyoming	Wyoming toad <i>Bufo hemiophrys baxteri</i>	Agriculture, unknown causes ³
Southeastern New Mexico, western Texas	Dune lizard <i>Sceloporus arenicolus</i>	Oak control ⁴
Illinois, Iowa, Missouri	Yellow mud turtle <i>Kinosternon flavescens</i> (northeastern populations)	Habitat loss ⁵

¹ Morafka 1982

² A.H. Price (personal communication)

³ Corn 1994

⁴ H.L. Snell (personal communication)

⁵ Dodd 1983

Illinois), but the only forms that appear to be genuinely threatened with extinction are the few relicts with restricted distributions resulting from post-Pleistocene climatic changes.

The Bolson tortoise (*Gopherus flavomarginatus*) of the Mexican plateau is a grassland species with special problems (Morafka 1982). Human predation for food, exacerbated by recent colonization of large ranches of northern Mexico, has reduced the tortoise to small scattered enclaves. Fortunately, efforts to protect the tortoise on an 18,000 ha private ranch in northern Mexico have apparently been successful, and it is hoped that the same sort of agreement can be obtained from other ranches that harbor tortoises (Anonymous 1994). Measures agreed to by the ranch's owners include modification of grazing rotations to protect hatchling tortoises and hiring of a part-time warden to patrol for poachers.

Habitat destruction and possibly pesticide use is responsible for the endangerment of other grassland species with limited ranges. The black-spotted newt (*Notophthalmus meridionalis*) of the lower Rio Grande (Río Bravo) Valley of Texas and

Mexico is an imperiled grassland species. Agriculture has destroyed many of the temporary ponds used by newts in the valley floor, but it still may be fairly common on large cattle ranches that are inaccessible for biological surveys (A.H. Price personal communication). The Wyoming toad (*Bufo hemiophrys baxteri*) did not breed in the wild from 1992-1994, but some adults were still present in 1994 (Corn 1994 personal communication). Pesticide use has been suggested as a reason for their demise, but evidence is lacking. Northeastern relict populations of the yellow mud turtle (*Kinosternon flavescens*) in Illinois and Iowa have been reduced to a few widely scattered ponds because of habitat destruction (Dodd 1983).

In the grasslands of the northern United States and southern Canada (Alberta, Colorado, the Dakotas, Iowa, Saskatchewan, Utah, Wyoming), there is a conservation issue of special concern. Western rattlesnakes, bullsnakes, racers (*Coluber constrictor*, *Masticophis taeniatus*), common garter snakes (*Thamnophis sirtalis*), and a few other snakes congregate in large numbers at widely scattered den sites in rocky outcrops or prairie dog towns

(Woodbury 1951). Here they are susceptible to decimation by killing, commercial exploitation, and prairie dog control efforts. Anecdotes of killing hundreds and even thousands of rattlesnakes in one year at a single den site were common in the past (Klauber 1972). There is no indication that any of these species are especially endangered, but their denning needs make them vulnerable to future overexploitation or decimation.

MANAGEMENT CONSIDERATIONS

Many species depend on either temporary or permanent bodies of water, and most forms that have suffered declines are associated with water or riparian vegetation. For this reason, water provision will often have the greatest immediate effect on local herpetofaunal species richness, especially on drier western and southwestern ranges.

Intensive grazing, grass plantings, brush removal, prairie dog control, off-road vehicle use, and other activities that reduce habitat complexity usually decrease the number of species of reptiles and amphibians. Management techniques that open up grasslands or encourage scattered trees and shrubs should have the most value in dense, well-watered eastern and northern areas. The greatest research needs are those directed at the effects of water provision, burning, grass planting, and overgrazing on various segments of the herpetofauna.

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